AI Department's Analysis of Data Generated by the Spandan Watch

AI-assisted physical-state and focus-monitoring analysis of Spandan Watch telemetry
20 October 2026

1. Key finding

The high "stress" figure is almost entirely a low-SpO_2 effect, not real stress

The report shows that 56.2% of all test observations were classified as **Stress**. When we look at *why* these readings were labelled as stress, we see that **about 99.99% of the stress labels are caused by low SpO_2 (blood-oxygen) readings alone**. Only about **0.01%** of the stress time comes from a raised heart rate, which is the only real sign of physical stress load in this data.

In simple terms: the watch is repeatedly reporting oxygen levels that are too low to be believable for officers who are sitting or standing still and whose heart rate is normal. This pattern points to a **watch-fit or sensor problem and possibly corrupted SpO_2 data**, rather than to officers who are truly under stress.

Once these doubtful SpO_2 readings are set aside, the stress share falls to roughly **0.01%**. That is a far more logical figure for officers whose heart rate is normal, so the 56.2% headline should not be read as real stress.

Suggested way forward

In view of this, it would be sensible for us to recalibrate the watch. A few simple steps may help:

- **Recalibrate the watch** and make sure it is worn snugly and aligned properly on the wrist. A loose or rotated watch can give wrong SpO_2 readings, and poor fit is the most likely cause of the pattern seen here.
- **An SpO_2 checkup** against a reliable fingertip pulse oximeter on the same officer at the same time would help confirm whether the watch readings are correct.
- **Once the SpO_2 readings are corrected**, the predictions can be re-run so that the stress, activity and normal figures can be confirmed.

2. How we arrived at this finding

The stress label is not a direct measurement of stress. It is assigned by a simple rule: a reading is called **Stress** when the officer is hardly moving and either the heart rate is high (above 90 bpm) or the SpO_2 is low (below 91%). The models learn this rule, so whatever drives the label in the data also drives the model's predictions.

Looking at what actually triggers the label, the picture is clear:

- **Heart rate is normal.** For PSI Kavita Mistry the median heart rate is 87 bpm, below the 90 bpm threshold. None of the stress labels come from heart rate alone, or from heart rate and SpO_2 together.
- **Low SpO_2 is the only trigger.** The median SpO_2 is 89%, with 51% of readings below 90% and 21% below 85%. All 4,144 stress-labelled readings (100%) come from SpO_2 below 91% alone.
- **The officer is mostly still.** About 52% of readings are stationary (under 1 km/h), so the rule treats the low oxygen value as "unexplained by movement" and labels it stress.
- **The dashboard agrees.** Its own AI assessment warns of a possible sensor-fit issue rather than a physiological problem, and suggests checking the fit or running a calibration check.

A healthy officer does not sit at 85–90% blood oxygen for hours while the heart rate stays normal. Such readings are more likely a sign of poor contact or a faulty sensor. That is why the SpO_2 data needs to be

checked before any of the stress figures in this report are used.

Item	What the data shows
Share of stress labels caused by low SpO_2 alone	About 99.99% (100% for the worked example, PSI Kavita Mistry)
Share of stress caused by raised heart rate	About 0.01%
Stress share after removing doubtful SpO_2 readings	About 0.01% (a logical figure for normal heart rates)
Most likely cause of the pattern	Watch fit/alignment or corrupted SpO_2 data

Table 1. Summary of the stress-label finding

3. Data and models used

The AI Department analysed the chronological held-out test data generated by the Spandan Watch. The analysis covered 15,113 telemetry observations from 29 officers and 1,385 focus-monitoring windows. The overall distribution identified by the models is given in Table 2. **Please read the stress figure together with Section 1: it mainly reflects low SpO_2 readings and should not be taken as real stress.**

Measure	Share
Stress (physical state)	56.2%
Physical activity (physical state)	4.2%
Normal state (physical state)	39.6%
Distracted focus windows	23.1%

Table 2. Overall distribution of model outputs on the test set

The analysis used the repository's best real-duty-data models, not the synthetic-only or lower-performing comparison models. Both were applied to the chronological test set reserved for evaluation.

Analytical function	Model used
Physical-state prediction: Stress, Physical Activity or Normal	Best real-duty physical-state model
Focus prediction: Focused or Distracted	Best real-duty focus-monitoring model

Table 3. Models applied

All results are model-generated screening estimates. They are not a clinical diagnosis and not a confirmed measurement of psychological stress. Sensor artefacts, missing-sensor imputation, duty-specific exertion, environmental conditions and the rule-based nature of the labels can all affect them.

4. Officer-level findings

The complete officer-level analysis is provided in officer_summary.csv, which covers all 29 officers. Table 4 presents the six officers with the largest test samples; Figure 1 compares all officers.

Officer	Test rows	Stress (%)	Activity (%)	Normal (%)	Distracted windows (%)
PSI Kavita Mistry	2,344	56.1	7.6	36.3	34.1
PI Bharat Rathod	2,211	60.6	0.6	38.8	12.0
ASI Meera Makwana	1,650	80.1	1.3	18.6	38.4
PSI Kiran Mistry	1,385	45.1	1.8	53.1	10.5
PI Nilesh Thakor	1,205	46.8	3.7	49.5	13.9
Constable Dinesh Chaudhary	913	55.8	1.5	42.7	26.7

Table 4. Officer-level results for the six largest test samples

Stress exposure looks high for all six officers (45–80% of test rows). Because this stress is driven mainly by low SpO_2, the follow-up for each officer is the same: check the watch fit and alignment and verify SpO_2 with a pulse oximeter. ASI Meera Makwana shows the highest stress share (80.1%) and also one of the higher distracted-window shares (38.4%).

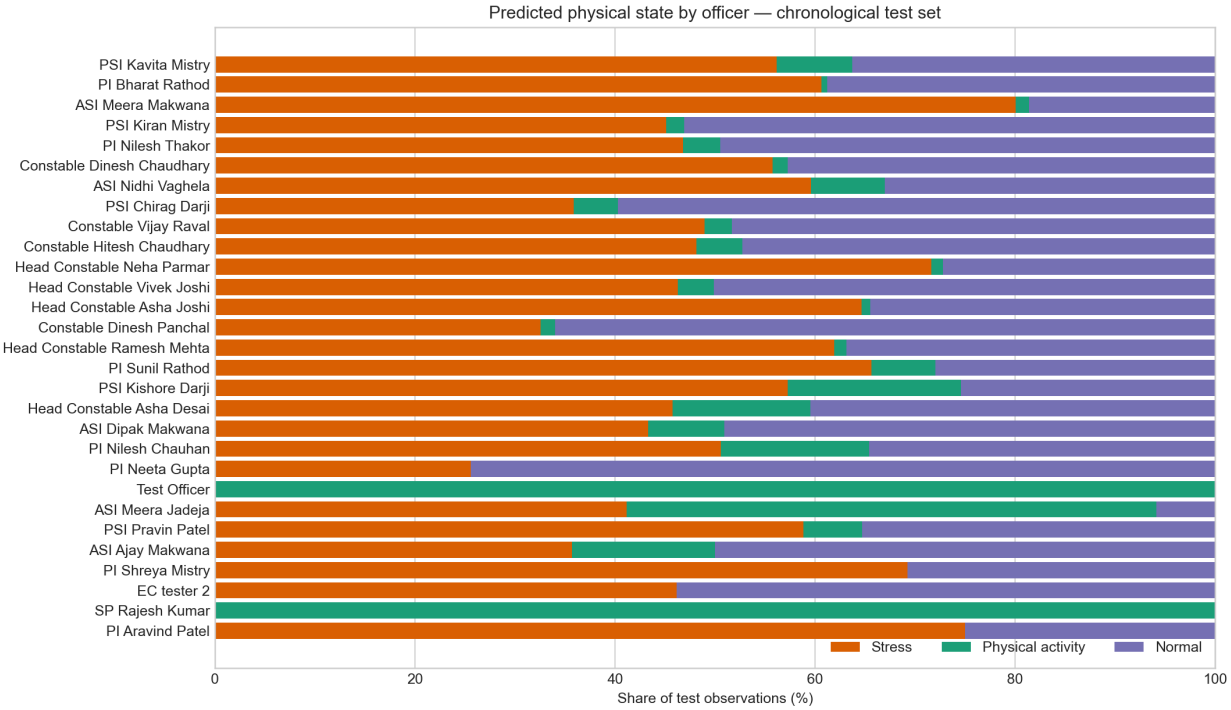


Figure 1. Predicted physical state by officer (chronological test set).

Figure 1 shows that stress is the dominant predicted state for most officers, with large differences between individuals. Two entries (Test Officer and SP Rajesh Kumar) are 100% physical activity and one (EC tester 2) has a single-class profile; these appear to be test or low-volume accounts and should be read with caution rather than as duty findings.

Focus monitoring and overall state mix

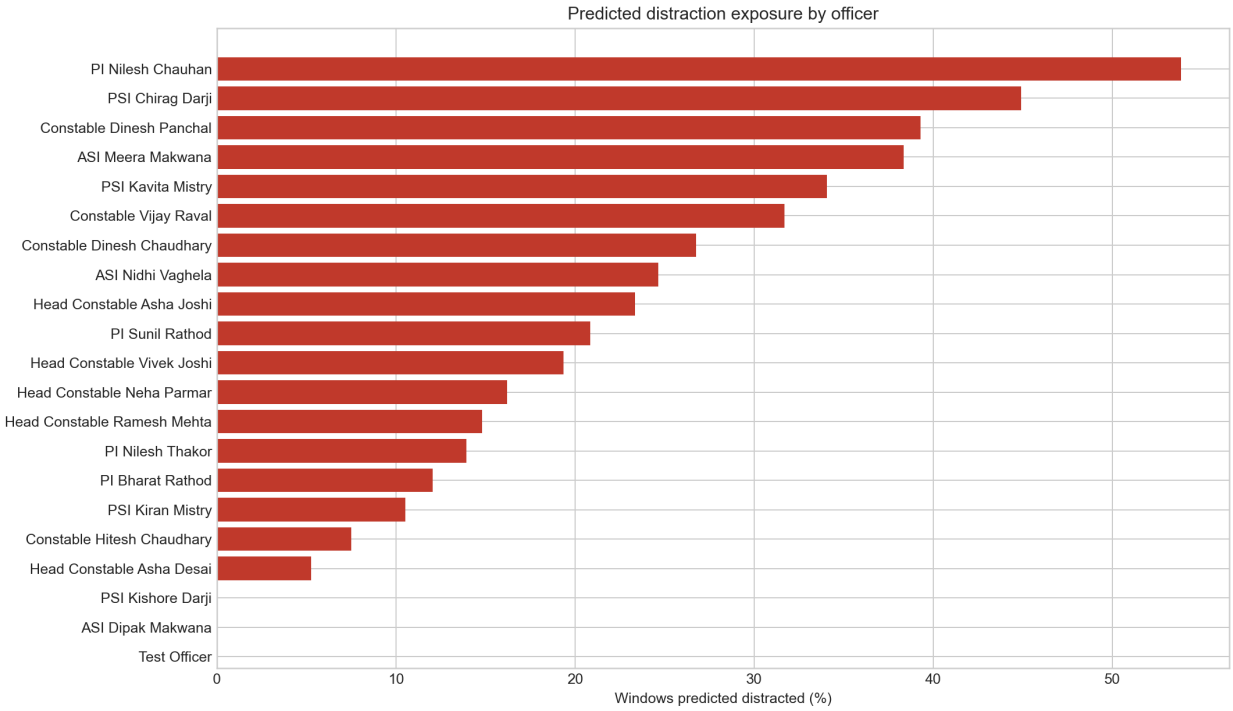


Figure 2. Predicted distraction exposure by officer (share of focus windows predicted distracted).

Distraction exposure varies widely. PI Nilesh Chauhan has the highest share (just over half of windows), followed by PSI Chirag Darji and Constable Dinesh Panchal. Only 20 of the 29 officers appear in this chart, and three of those show no distracted windows. Among the officers in Table 4, PSI Kavita Mistry (34.1%) and ASI Meera Makwana (38.4%) are the highest.

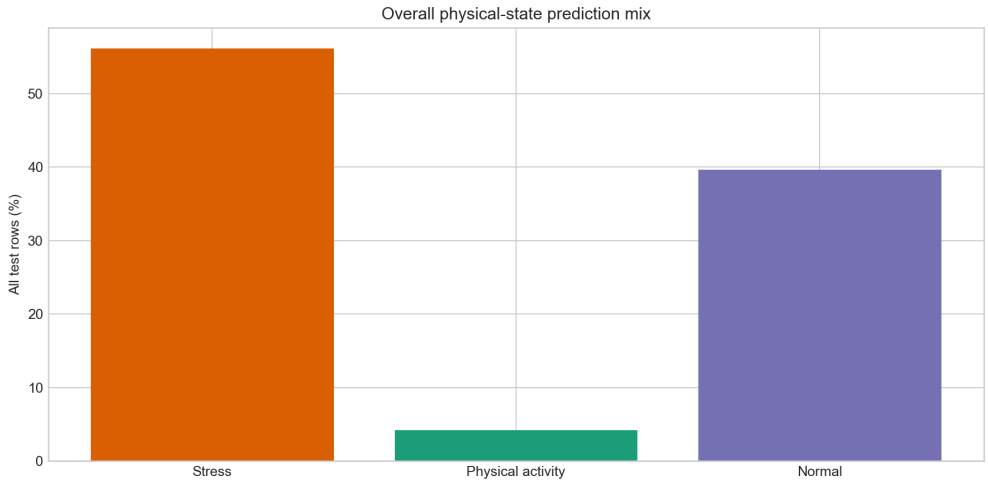


Figure 3. Overall physical-state prediction mix across all test rows (Stress 56.2%, Physical activity 4.2%, Normal 39.6%). The stress bar is largely a low-SpO_2 effect (see Section 1).

5. Dashboard screenshot evidence

The Police Personnel Health Analytics dashboard was used to check the reasoning above at officer level, with PSI Kavita Mistry as the worked example. The dashboard displays the labelled *training split*, whereas Sections 1–4 report model predictions on the held-out *test set*. The two are therefore compared qualitatively (same signals and thresholds), not number for number.

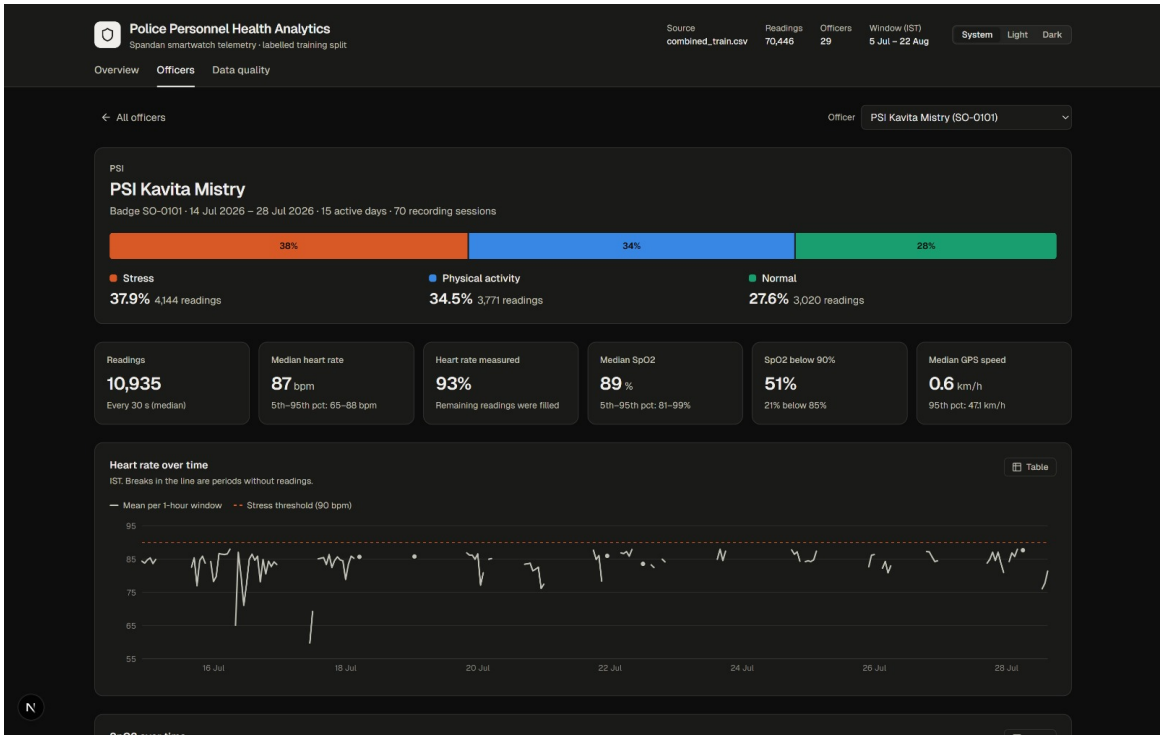


Figure 4. Officer-level overview: label mix, key vital-sign statistics, and heart-rate trend against the 90 bpm stress threshold.

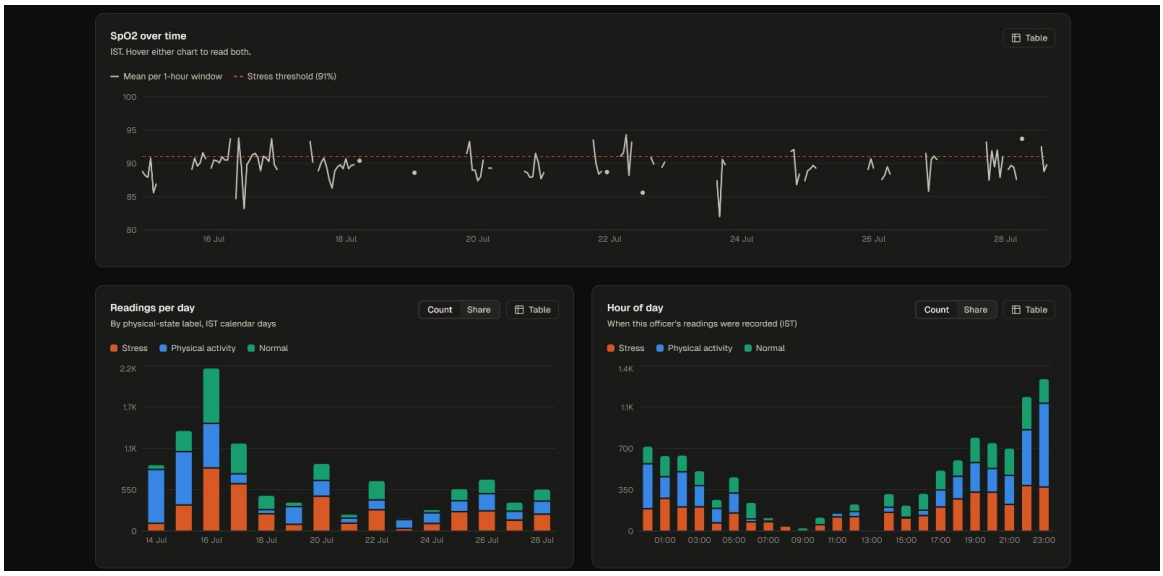


Figure 5. SpO_2 trend against the 91% threshold, readings per day, and hour-of-day distribution.

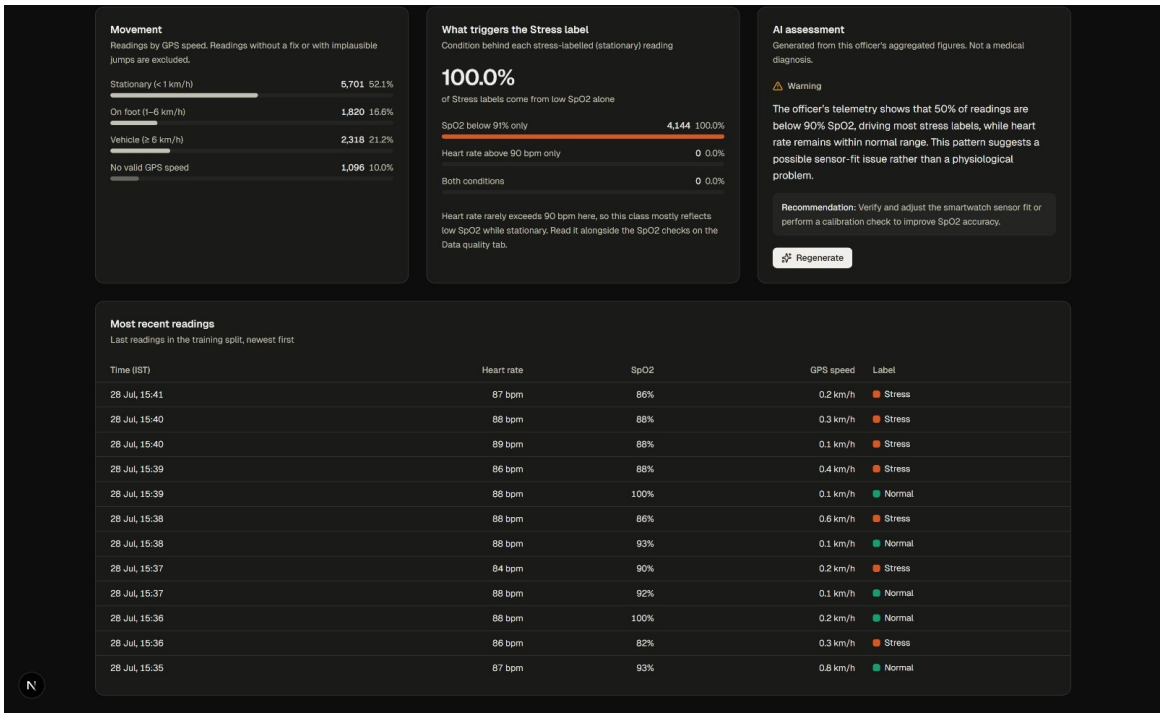


Figure 6. Movement profile, stress-label triggers, AI assessment, and the most recent telemetry readings.

5.1 What the dashboard shows

- **Heart rate is not the trigger.** Median heart rate is 87 bpm (5th–95th percentile 65–88 bpm), below the 90 bpm threshold. None of the stress labels arise from heart rate alone or from both conditions.
- **Low SpO_2 drives the label.** Median SpO_2 is 89%, with 51% of readings below 90% and 21% below 85%. All 4,144 stress-labelled readings (100%) come from SpO_2 below 91% alone.
- **Movement is low when stress is labelled.** About 52% of readings are stationary (under 1 km/h), consistent with the rule that stress is flagged when load is not explained by movement.
- **The dashboard's own AI assessment flags a sensor-fit concern.** It suggests a possible sensor-fit issue rather than a physiological problem and recommends checking fit or running a calibration check.

5.2 Reconciling the dashboard with the test-set results

Table 5 sets out why the dashboard and report figures differ.

	Dashboard (Figures 4–6)	Report analysis (Sections 1–4)
Data source	combined_train.csv, labelled training split	Chronological held-out test set
Volume	70,446 readings; 29 officers	15,113 observations; 29 officers
Stress, activity and normal labels	Project labels (heuristic rules)	Model predictions
PSI Kavita Mistry (stress / activity / normal)	37.9% / 34.5% / 27.6%	56.1% / 7.6% / 36.3%

Table 5. Dashboard and report analysis compared

The figures differ because the data differ: the dashboard summarises 70,446 labelled training readings, whereas the report analyses 15,113 held-out test observations with model predictions. What the two share is the mechanism. In both, stress is assigned to stationary readings with low SpO_2 or elevated heart rate, and in the data it is the low SpO_2 that does almost all of the work. This is why the elevated stress share must be validated against sensor quality before it is treated as a physiological signal.

6. Conclusion

The Spandan Watch telemetry shows the potential of AI-assisted monitoring for identifying patterns of cardiovascular load, physical activity and possible distraction. However, the high stress share found in this analysis is almost entirely caused by low SpO_2 readings (about 99.99%), with only about 0.01% coming from a raised heart rate. This points to watch fit or SpO_2 data problems rather than real stress.

Once the watch alignment and SpO_2 readings have been checked and corrected, the predictions can be re-run and confirmed. The results should be used for early screening and timely review only. They are not a substitute for clinical assessment, confirmed incident evidence, or professional judgement.